



Universidade do Minho
Escola de Engenharia
Departamento de Informática

Knime

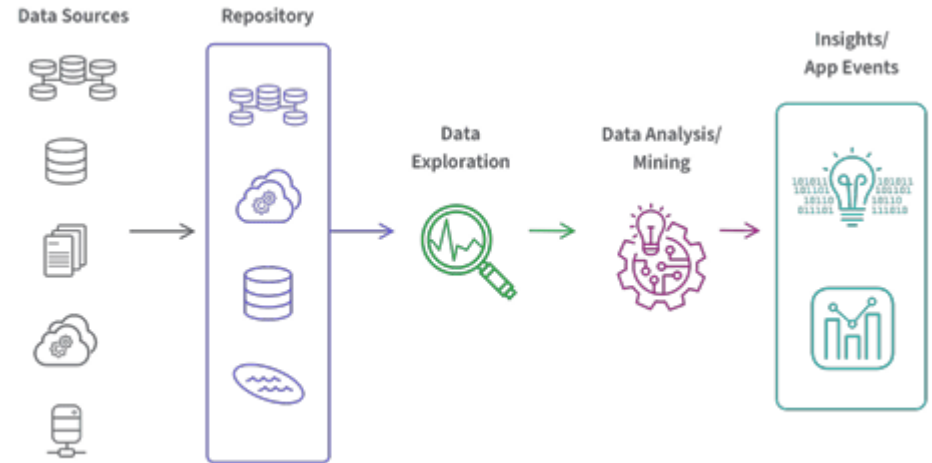
Pre-Processing

MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E INFORMÁTICA
MESTRADO integrado EM ENGENHARIA DE TELECOMUNICAÇÕES E
INFORMÁTICA

Inteligência Artificial para as Telecomunicações
2025/2026

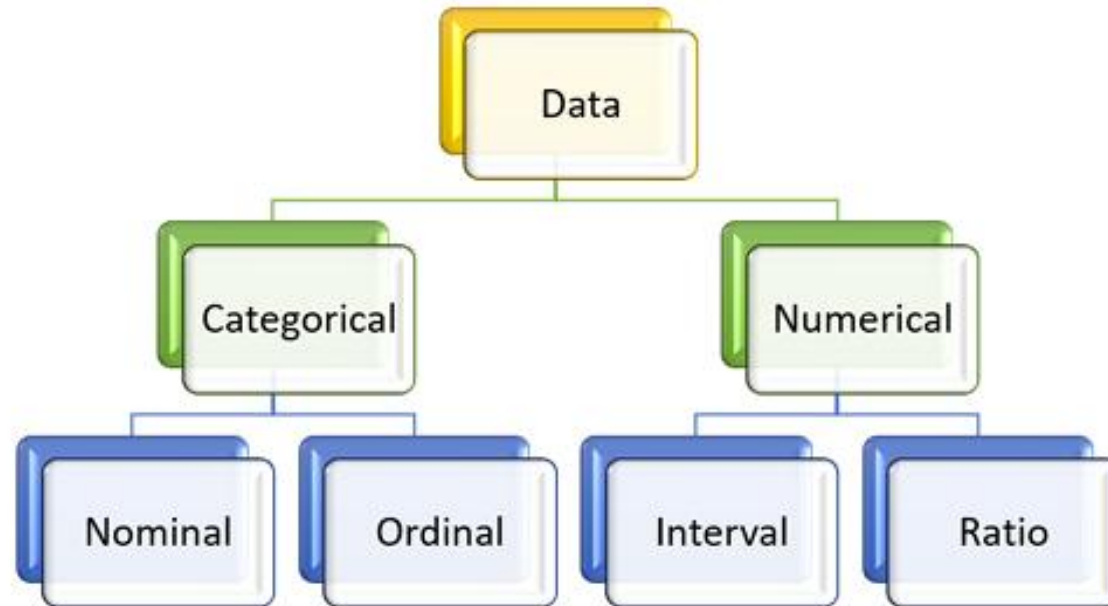
Data Exploration

- **Examine and visualize** dataset to understand its main characteristics, identify **patterns, anomalies and relationships** within the data.
- Useful to gain **insights** for subsequent analysis and modelling steps.



Unprepared data can be:

- **Incomplete** (lack of values in some attribute, lack of some attributes)
- **Uncleaned** (identify impossible values, such as an age of 321 years old)
- **Inconsistent** (age of 35 but birthdate is May 31 1961)



Qualitative Data:

- **Nominal** – names or labels with no order (gender, colors, fruit names)
- **Categorical** – limited or fixed number of possible values often grouped into categories or labels (gender, zip code)
- **Ordinal** – values have a meaningful ordered relationship, allowing for ranking or comparison (satisfaction rates: “poor”, “average”, “bad”; Education levels, likert scales)

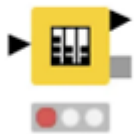
Quantitative Data:

- **Interval** – difference between values are meaningful and consistent but there is no true zero point, meaning ratios cannot be calculated (Celsius, Fahrenheit)
- **Ratio** – difference between values are meaningful and consistent and there is no true zero point, meaning ratios can be calculated (height, weight)

- **Discretization/Enumeration** – converting continuous data into discrete intervals or categories.

Eg: Age. Instead of keeping each possible age as a unique value (22, 20, 18), discretization would group it into bins: 0-10, 11-20, 21-30, etc

Auto-Binner



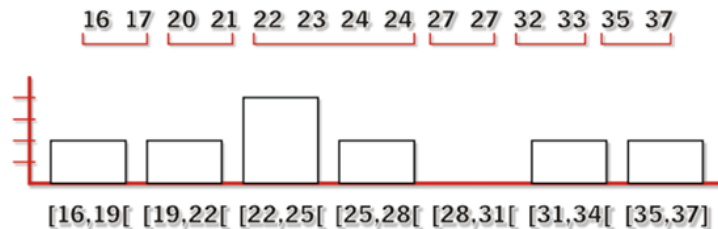
In Knime, the Auto-Binner node is used to automatically bin or discretize continuous data into categorical bins.

➤ Discretization/Enumeration

Equal-Height: the same number of data points, regardless of the width.

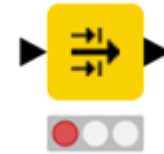


Equal-Width: intervals of equal size.



➤ Cleaning

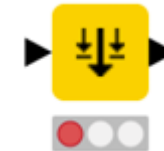
Remove missing data



Row filter node

-Complete Case Deletion: remove rows (entries)

-Column Deletion: Remove columns (features)



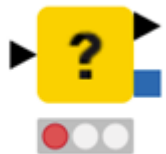
Column filter node

➤ Cleaning

Imputation

-Mean/Median/Mode amputation: replace missing values with the mean (for numerical data), median (if the data is skewed) or mode (for categorical data)

This works well for **small datasets** or cases where the feature is not critical, however, it can distort the data distribution, reducing variability and potentially biasing results.

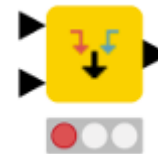
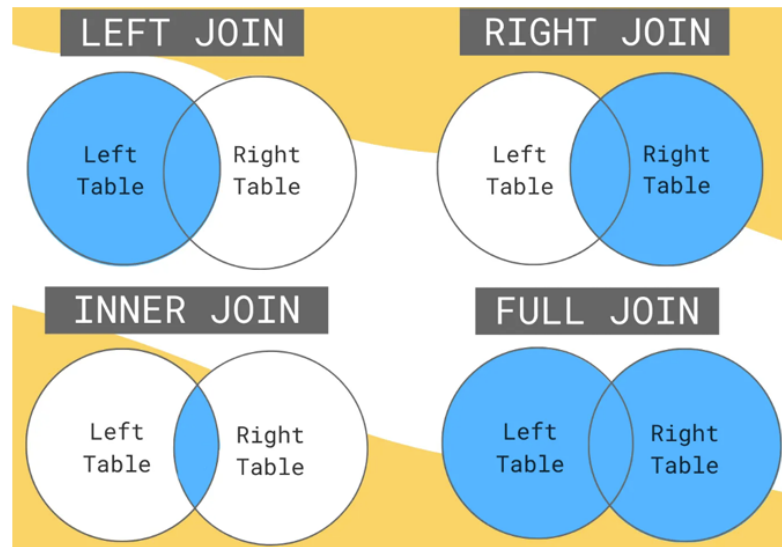


missing value node

➤ Integration

Joining two tables (datasets)

- **Inner Join:** Only includes rows that have matching keys in both tables.
- **Left Join:** Includes all rows from the left table and matching rows from the right table; unmatched rows in the right table will have null values.
- **Right Join:** Includes all rows from the right table and matching rows from the left table; unmatched rows in the left table will have null values.
- **Full Join:** Combines rows from both tables, including all unmatched rows from both sides, with null values where there are no matches.



Joiner node

➤ Transformation

Normalization

The process of scaling numerical values to a **standard range**, usually between **0 and 1** or **-1 and 1**. Normalization is important when features have different units or scales, ensuring that no single feature dominates others during analysis or model training.

Normalization methods:

Min-Max Scaling: rescales the data to a range of $[0, 1]$

Z-Score: standardizes data to have a mean of 0 and a standard deviation of 1.



Normalizer node

➤ Transformation

Encoding Categorical Variables

Converts **categorical variables** into **numerical** representations.

Label Encoding: Assigns each category a unique integer. For example, "Red" = 0, "Blue" = 1, "Green" = 2.

One-Hot Encoding: Creates binary columns for each category, representing the presence of a category as 1 or 0.

Label Encoding

Food Name	Categorical #	Calories
Apple	1	95
Chicken	2	231
Broccoli	3	50



One Hot Encoding

Apple	Chicken	Broccoli	Calories
1	0	0	95
0	1	0	231
0	0	1	50



Rule Engine node

➤ Transformation

Encoding Categorical Variables



Rule Engine node

The Rule Engine Node allows you to define rules for transforming data based on specific conditions, making it suitable for various encoding tasks.

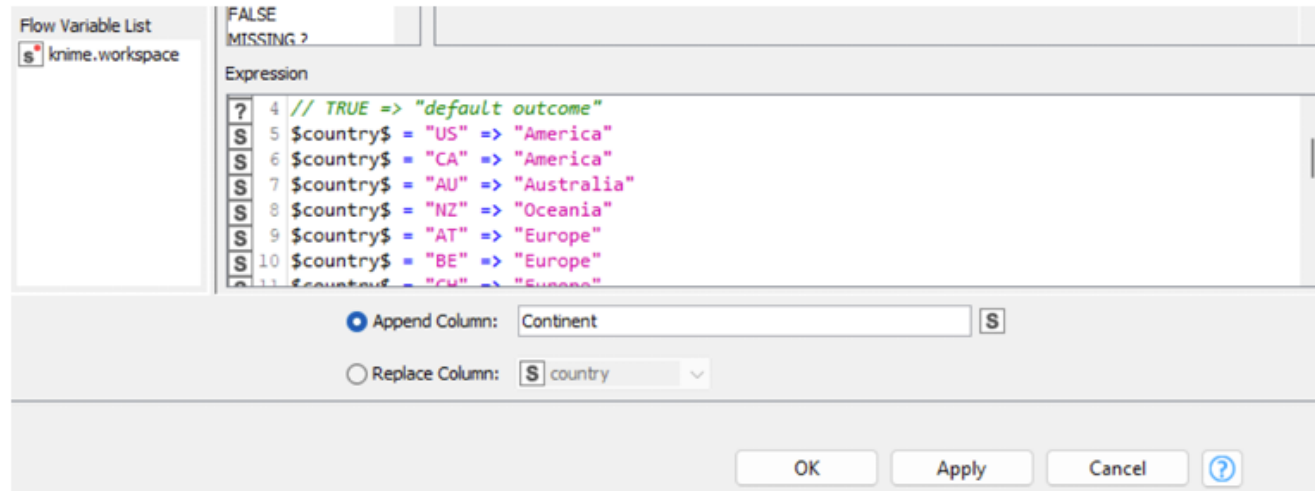
The expressions below illustrate an example that encodes the "Favorite Fruit" column into numerical values:

```
$Favorite Fruit$ = "Apple" => 0  
$Favorite Fruit$ = "Banana" => 1  
$Favorite Fruit$ = "Orange" => 2
```

➤ Transformation

Feature Engineering

The process of **creating new features or modifying** existing ones to improve the performance of machine learning models. For example, extracting features like year, month, day, weekday, etc., from datetime objects. The example below maps the country acronyms values to their corresponding continent.

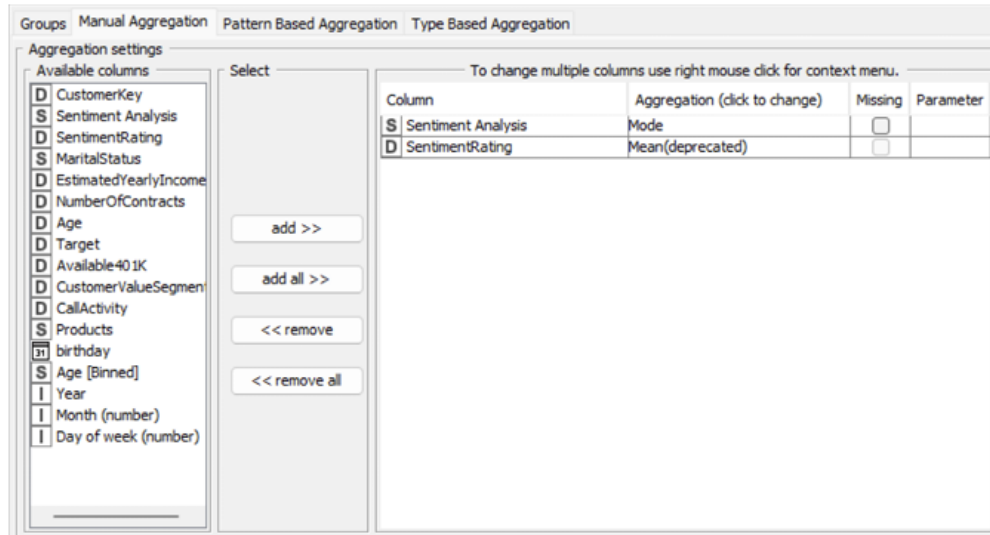


Rule Engine node

➤ Transformation

Aggregation

Summarizing data by grouping and applying aggregation functions such as **sum**, **average**, or **count**. Aggregation is useful for reducing data volume and for generating summary statistics for analysis.



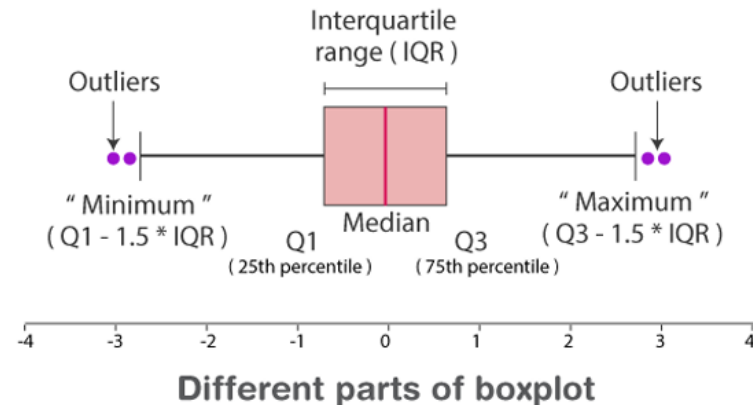
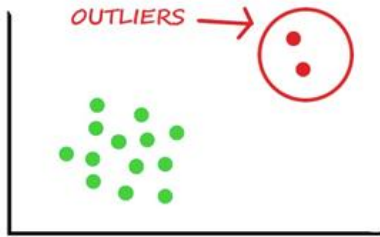
Group By node

➤ Transformation

Outlier Detection

An outlier is defined as a data point that **differs significantly** from other observations in the dataset, and identifying these points is essential for ensuring data quality.

A widely used technique for identifying outliers is through the visualization of **Box Plots**.



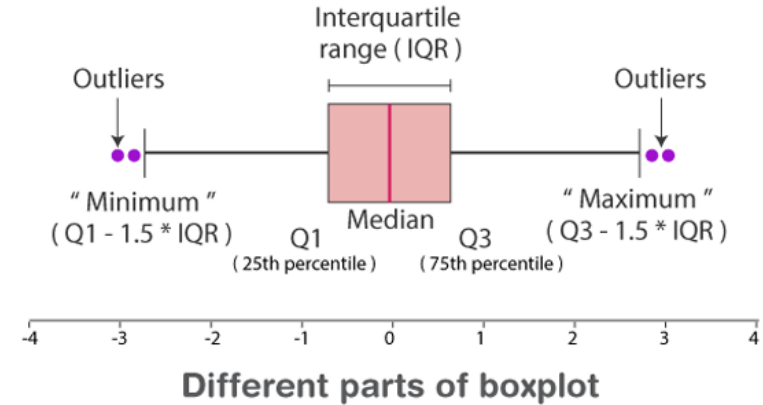
➤ Interpreting a Boxplot

-The line splitting the box in two represents the median value. This shows that 50% of the data lies on the left hand side of the median value and 50% lies on the right hand side.

– The left edge of the box represents the lower quartile; it shows the value at which the first 25% of the data falls up to.

– The right edge of the box shows the upper quartile; it shows that 25% of the data lies to the right of the upper quartile value.

– The values at which the horizontal lines stop at are the values of the upper and lower values of the data. The single points on the diagram show the outliers.

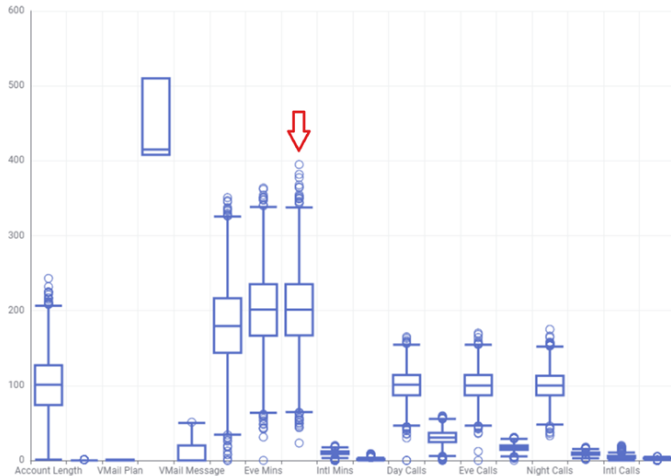


➤ Transformation

Outlier Detection

In KNIME, the node that allows to visualize this plot is called **BoxPlot**.

The example below is an output generated by the node in question when you get to see several outliers.



The Numeric Outliers node allows handling these outliers.



Numeric Outliers node

➤ Transformation

Outlier Detection

- **Remove Outliers:** Just take out the outlier data points from your dataset. Use this if the outliers are mistakes or don't represent the data you care about.
- **Cap or Limit Outliers:** Change the outlier values to a maximum or minimum value within a reasonable range. Apply this when you want to keep the data but reduce the impact of extreme values.
- **Transform Data:** Change the data with mathematical formulas to make it easier to handle. Use this when outliers are important but you need to reduce their influence.

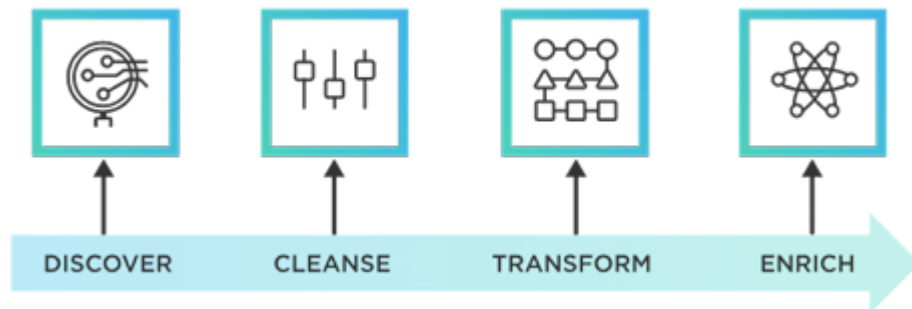
The screenshot displays the SAS Outlier Detection interface. The 'Outlier Selection' tab is active, with 'Manual Selection' chosen. The 'Exclude' list (highlighted with a red box) contains 'Int'l Plan', 'VMail Plan', and 'Area Code'. The 'Include' list (highlighted with a green box) contains 'Account Length', 'VMail Message', 'Day Mins', 'Eve Mins', 'Night Mins', 'Int'l Mins', 'CustServ Calls', 'Day Calls', 'Day Charge', 'Eve Calls', and 'Night Calls'. The 'General Settings' section shows 'Interquartile range multiplier (k)' set to 1.5 and 'Full data estimate using' set to 'R_4'. The 'Outlier Treatment' section shows 'Apply to' set to 'All outliers' and 'Treatment option' set to 'Remove outlier rows' (both highlighted with a red box). The 'Replacement strategy' is set to 'Missing values'.

➤ Transformation

Reduction

Data Reduction aims to obtain a reduced representation of the data volume, but producing the same (or almost the same) analytical results.

- **Remove** attributes that are irrelevant, redundant or uninteresting for the analysis.
- Apply **compression or transformation** techniques to compress the representation of original data.
- Reduce the number of values per attribute through **discretization** and generalization of concepts.





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